11. Sequences and Series

11.2. Series

11.2a Learning Objectives
- Understand what a series means in relation to sequences
- Identify types of series (geometric and harmonic)
- Determine when a series converges or diverges

Suppose we have a sequence, $(a_n)_{n=1}^{\infty}$, and we want to add up all the terms. How do we sum an infinite number of values? (the short story is: very carefully, and one term at a time)

Definition 11.2.1 **Series**

Given a sequence of numbers (a_n), and expression of the form

$$a_1 + a_2 + a_3 + a_4 + ... + a_n + ...$$

is an **infinite series**.

The sequence (s_n) is defined by

$$\begin{aligned} s_1 &= a_1 \\ s_2 &= a_1 + a_2 \\ s_3 &= a_1 + a_2 + a_3 \\ &\vdots \\ s_n &= a_1 + a_2 + a_3 + ... + a_n = \sum_{k=1}^{n} a_k \\ &\vdots \end{aligned}$$

is the **sequence of partial sums** of the series, the number s_n being the $\boldsymbol{n^{\text{th}}}$ **partial sum**.

If the new sequence, (s_n), of partial sums converges to a limit L, we say that the *series* **converges** and that its sum is L. In this case, we also write,

$$a_1 + a_2 + ... + a_n + ... = \lim_{n\to\infty} s_n = \lim_{n\to\infty} \sum_{k=1}^{n} a_k = \sum_{k=1}^{\infty} a_k = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

Example 11.2.1

Determine the convergence of the series $\sum_{k=1}^{\infty} (-1)^k$ using partial sums.

Solution

The *sequence* this series comes from is $a_1 = -1, a_2 = 1, a_3 = -1, a_4 = 1, ...$ This means the *sequence of partial sums* for the series is

$$
\begin{aligned}
s_1 &= -1 \\
s_2 &= -1 + 1 = 0 \\
s_3 &= -1 + 1 - 1 = -1 \\
s_4 &= -1 + 1 - 1 + 1 = 0 \\
&\vdots
\end{aligned}
$$

The sequence of partial sums, (s_n), diverges, so the series, $\sum_{k=1}^{\infty} (-1)^k$, diverges.

Example 11.2.2

Determine the convergence of the **telescoping series** $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ using partial sums.

Solution

We can rewrite $\frac{1}{n(n+1)}$ with partial fractions:

$$
\begin{aligned}
\frac{1}{n(n+1)} &= \frac{A}{n} + \frac{B}{n+1} \\
1 &= A(n+1) + Bn
\end{aligned}
$$

So the system we get is $A + B = 0$ and $A = 1$, so $B = -1$. Then

$$
\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}
$$

This will help us write the sequence of partial sums. The nth partial sum is then

$$
\begin{aligned}
s_n &= \left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n} - \frac{1}{n+1}\right) \\
&= \frac{1}{1} + \left(-\frac{1}{2} + \frac{1}{2}\right) + \left(-\frac{1}{3} + \frac{1}{3}\right) + \ldots + \left(-\frac{1}{n} + \frac{1}{n}\right) - \frac{1}{n+1} \\
&= 1 - \frac{1}{n+1}.
\end{aligned}
$$

Now we have a nice way to write the sequence of partial sums. The sequence of partial sums $(s_n)_{n=1}^{\infty} = \left(1 - \frac{1}{n+1}\right)_{n=1}^{\infty}$ converges since

$$
\lim_{n\to\infty} s_n = \lim_{n\to\infty} 1 - \frac{1}{n+1} = 1
$$

So the series converges to 1:

$$
\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1
$$

Example 11.2.3

Determine the convergence of the **harmonic series** $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ using partial sums.

Solution

If n is a power of 2, we get

$$
\begin{aligned}
s_1 &= 1 = 1 + 0\left(\frac{1}{2}\right) \\
s_2 &= 1 + \frac{1}{2} = 1 + 1\left(\frac{1}{2}\right) \\
s_4 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} > 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 + 2\left(\frac{1}{2}\right) \\
s_8 &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} \\
&> 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) \\
&= 1 + 3\left(\frac{1}{2}\right)
\end{aligned}
$$

This pattern continues, showing that $s_{2^n} \geq 1 + \frac{n}{2}$, so the sequence of partial sums is unbounded so it diverges. Thus, the harmonic series diverges.

11.2a Section Summary:

- We defined a series using the sequences of partial sums.
- We used the sequence of partial sums of series to discuss the convergence or divergence of some series.
- We looked at **telescoping series** and showed **the harmonic series** diverges.